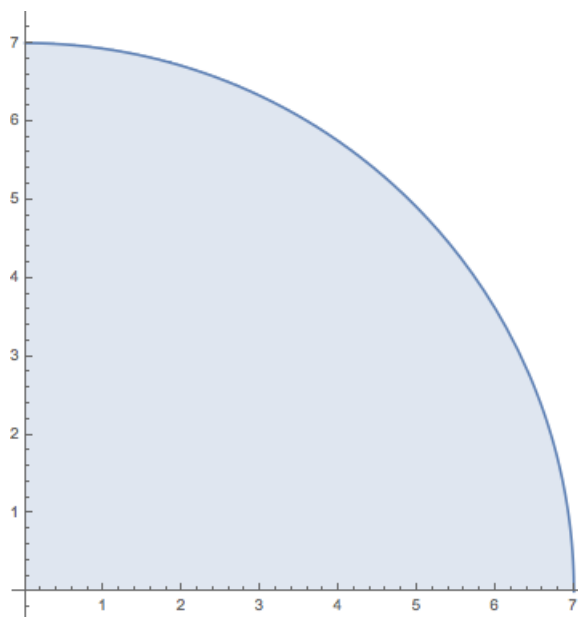


Problem 7

Problem 7

Problem 1 A part of a circle is shown in the figure.



- (a) If we express the area of the shaded region in the figure as the limit of Riemann sums

$$A = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x,$$

find the function f and the interval $[a, b]$.

- (b) Compute the limit.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{49 - (x_k^*)^2} \cdot \frac{7}{n}$$